TOUR CEYLON: AN INTEGRATED AI-DRIVEN PLATFORM FOR TOURIST SAFETY, ITINERARY PLANNING, WELLNESS MATCHING AND CROWD PREDICTION IN SRI LANKA

COMMON INTEGRATED SOLUTION REPORT

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Report submitted in partial fulfilment of the requirements for the

Bachelor of Science (Honours) in Information Technology

Specialising in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

International tourists in Sri Lanka have no single, trustworthy digital service that plans a feasible trip from registered places, indicates how busy sites will be, matches them to suitable wellness care and brings help quickly in an emergency. Information is fragmented, generic AI planners are not grounded in local data, site statistics are annual totals and emergency channels are voice-based. This report presents Tour Ceylon, an integrated platform built by four students following design science research. An SOS system classifies a tourist's statement and routes a GPS-anchored alert to the nearest police station's dashboard; an itinerary generator turns free-text wishes into calendar-valid, timed plans from catalogues of 343 attractions and 921 accommodation units; a risk service predicts a daily crowd score for 197 sites; and a wellness component matches tourists to 83 verified Ayurveda centres and retreats in five languages. The components share one login, one Flask back end and one PostgreSQL database, and each keeps a fail-safe behaviour when its model or the network fails. Evaluated against simple baselines under stated designs, the crowd model reduced the error of the best baseline on ten unseen sites by about 40% (MAE 0.1232 against 0.2057), wellness matching raised top-match quality from 3.41 to 4.61 out of 5, and SOS responder routing reached about 0.81 accuracy. Honest evaluation also exposed weaknesses: the served fourteen-class SOS model scored below keyword rules, near-perfect itinerary scores reflect synthetic labels, and no component has yet been validated with real tourists. The work shows that a grounded, integrated tourist companion is feasible, and identifies real-data collection and integrated user testing as the steps needed to show real-world effectiveness.

Keywords: smart tourism, Random Forest, integrated platform, tourist safety, Sri Lanka

ACKNOWLEDGEMENTS

We sincerely thank our supervisor, Mr. S. M. B. Harshanath, and our co-supervisor, Prof. Samantha Rajapaksha, for their guidance and critical feedback throughout this project, particularly on evaluation design and on reporting machine-learning results honestly. Their feedback at the progress presentations shaped the reproducible pipelines and the evaluation standard described in this report.

We are grateful to the Ayurvedic practitioner who acted as domain expert for the wellness component and provided anonymised real patient cases and the condition, treatment, duration and cost structure; to the officials and tourists who gave their views on the SOS prototype and proposal; and to the peers who tested the itinerary wizard and the integrated platform.

We thank the Department of Information Technology and the Faculty of Computing of the Sri Lanka Institute of Information Technology for the facilities and support provided. Official statistics from the Sri Lanka Tourism Development Authority informed the problem framing. No external funding was received for this work. Generative AI tools assisted with drafting and structuring parts of the text, as disclosed in the individual theses; the team remains responsible for all content.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the ROC Curve
BB / HB / FB	Bed and Breakfast / Half Board / Full Board
CV	Cross-Validation
DSR	Design Science Research
E2E	End-to-End
F1	Harmonic mean of precision and recall
FIT	Free Independent Traveller
GPS	Global Positioning System
JSON	JavaScript Object Notation
JWT	JSON Web Token
LKR	Sri Lankan Rupee
LOO	Leave-One-Out
MAE	Mean Absolute Error
ML	Machine Learning
NFR	Non-Functional Requirement
PDPA	Personal Data Protection Act, No. 9 of 2022 (Sri Lanka)
REST	Representational State Transfer
RF	Random Forest
ROC	Receiver Operating Characteristic
R^2	Coefficient of determination
SDG	Sustainable Development Goal
SLTDA	Sri Lanka Tourism Development Authority
SOS	Distress alert or emergency help request
SUS	System Usability Scale
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
UAT	User Acceptance Testing

1 INTRODUCTION AND PROJECT CONTEXT

This chapter introduces the domain in which the Tour Ceylon platform operates, the people it serves and the evidence that the common problem matters. It then aligns the project with the Sustainable Development Goals and explains how this common report relates to the four individual theses.

1.1 Background and context

Tourism is one of Sri Lanka's most important sources of foreign exchange and employment. The Sri Lanka Tourism Development Authority (SLTDA) recorded a record 2,362,521 international arrivals in 2025, an increase of 15.1% over the 2,053,465 arrivals of 2024 [1]. Spending has not kept pace with arrivals: the survey-based average expenditure per tourist per day was revised from USD 171.74 for January to July 2025 to USD 148.26 for August to December 2025 [2], and the Central Bank of Sri Lanka has noted that the recovery in arrivals has outpaced the recovery in tourism earnings [3]. National policy therefore emphasises higher-quality experiences, longer stays and higher value per visitor, together with the safety and wellbeing of the visitors who come.

Visitors increasingly plan, travel and seek help through digital services. Smart tourism research describes destinations in which information and communication technologies personalise services for each visitor [4], and intelligent tourism systems have moved from simple information portals to recommender and decision-support tools [5], [6]. In practice, however, a foreign visitor to Sri Lanka still assembles a trip from many disconnected sources: map applications for directions, review and booking sites for hotels, blogs for routes, voice hotlines for emergencies and word of mouth for Ayurveda and meditation centres. None of these tools knows about the others, and none is grounded in Sri Lankan registers of attractions, sites or wellness providers.

Four kinds of need recur across the trip (Figure 1.1). Before the trip the visitor must decide where to go, how many nights to spend in each place and where to stay. Many visitors also choose a wellness programme: SLTDA recorded 12,977 visitors in 2025

whose main purpose was health or Ayurveda, and these visitors stayed 14.7 days on average, longer than any other group [1]. During the trip the visitor wants to avoid sites that are overcrowded on a given day, a concern that the overtourism literature links to the sustainability of destinations [7]. When something goes wrong (a theft, an accident, harassment, a medical emergency) the visitor must reach the right service quickly, explain where they are and describe what has happened, often in a second language [8]. Sri Lanka offers the 1912 tourism hotline, the 1997 tourist security line and the national 119 emergency number [9], [10], [11], but these channels are voice-based and are not linked to the visitor's location, profile or itinerary.

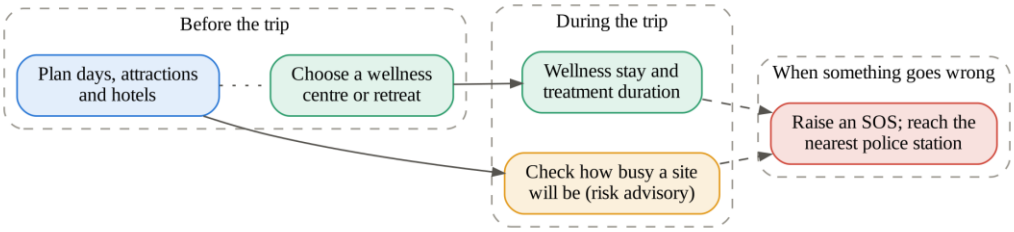


Figure 1.1: The tourist journey and the support provided by the four Tour Ceylon components

Tour Ceylon (research group R26-IT-152) responds to this situation with one platform built by four students. Each student owns one research component that also works as a standalone module, and the components are joined through a shared Flask back end, a shared PostgreSQL database, a single login and a common web front end. The platform therefore covers planning, crowd and risk awareness, emergency help and wellness matching within a single tourist identity.

1.2 Stakeholders and user needs

The stakeholders of the complete solution were identified from the project proposal, the literature, supervisor review and the stakeholder contact made by each component owner. They are summarised in Table 1.1. The primary users are international tourists, in particular free independent travellers (FITs) who plan their own trips and wellness tourists who stay for several weeks. Secondary users are the officers and businesses that respond to or serve these tourists.

Table 1.1: Stakeholders of the Tour Ceylon platform and their confirmed needs

Stakeholder	Role	Need confirmed through engagement or evidence	Components
International tourists (FITs)	Primary user	A realistic, timed plan with registered hotels and prices; awareness of busy sites; one place for the trip	Itinerary, Risk
Wellness tourists	Primary user	A suitable and safe Ayurveda centre or retreat for their condition, constitution, budget and language	Wellness
Tourists in distress	Primary user	One-tap help with automatic location, in their own words or voice, even without connectivity	SOS
Police and Tourism Police	Responder	Alerts that say who the tourist is, what happened and where, without lengthy clarification	SOS
Immigration officers	Oversight	Read-only view of consented events involving foreign visitors	SOS
SLTDA and site managers	Policy and operations	Spreading visitors across sites; a digital complement to the 1912 hotline	Risk, SOS
Hotels, tour operators, agents	Business partner	Qualified guests, duty-of-care support and itinerary building blocks	Itinerary, SOS
Ayurveda centres and retreats	Business partner	Visibility based on service quality; pre-qualified international guests	Wellness
Domain expert (Ayurvedic practitioner)	Knowledge provider	Correct condition–dosha–treatment mapping and realistic durations and costs	Wellness

Engagement differed by component. The SOS owner reports informal conversations with tourists in Colombo, Galle Face, Port City Colombo and Negombo, supervisor and stakeholder review of the SOS screens and police briefing format, and contact with Police Headquarters and immigration-related officials, including a proposal submitted to the Sri Lanka Police. The wellness owner worked with an Ayurvedic practitioner who supplied anonymised real patient cases, the condition–dosha–treatment mapping and the duration and cost structure. The itinerary owner validated the calendar rules and price presentation with the supervisor and with peers who used the wizard. The risk owner used published SLTDA site statistics. Across the team, three user needs were confirmed repeatedly: answers must be grounded in real, registered data rather than invented by a model; the tourist must not have to repeat

information that the platform already holds; and the platform must remain useful when an AI part fails or the network is unavailable.

User personas

Two personas guided the integrated design. *Anna*, a 34-year-old German independent traveller, arrives for ten days, wants beaches, culture and nature, and is considering a short Ayurveda programme for back pain. She needs a feasible day-by-day plan, a matched centre and a sense of which sites will be crowded on her travel dates. *Kenji*, a 52-year-old Japanese visitor travelling with a companion, is mainly concerned with safety; if his companion falls ill on a hiking trail he must be able to call for help in Japanese without knowing which number to dial or how to describe the location. The personas are illustrative and were not used as data.

1.3 Alignment with the Sustainable Development Goals

The project is aligned with five of the United Nations Sustainable Development Goals (SDGs) [12], as shown in Table 1.2. The contribution claimed is the intended contribution of a research prototype; no measured impact on any SDG indicator is claimed, and Section 7.5 evaluates possible negative effects.

Table 1.2: Alignment of the integrated solution with the Sustainable Development Goals

SDG and target	Intended contribution	Components
SDG 3 Good health and well-being	Faster medical emergency response; safer choice of traditional health care with safety flags and unverified providers marked	SOS, Wellness
SDG 8 Decent work and economic growth (target 8.9, sustainable tourism)	Directs visitors to registered attractions, hotels and wellness centres across many districts; supports longer, higher-value stays	Itinerary, Risk, Wellness
SDG 9 Industry, innovation and infrastructure	Reproducible, data-grounded digital services for the tourism sector, including digital public-safety infrastructure	All
SDG 11 Sustainable cities and communities	Safer tourist routes and crowd-aware visiting of heritage and nature sites	SOS, Risk
SDG 12 Responsible consumption and production (target 12.b)	Catalogue- and model-driven promotion of less-crowded and green sites without code changes	Itinerary, Risk

1.4 Purpose and scope of the common report

This common report presents the integrated problem and the integrated solution: the common problem and its four subproblems, the overall architecture, how the components work together, the shared implementation and project execution, the group-level evaluation and the commercial and responsible-innovation case for the complete platform. It deliberately does not repeat the detailed literature reviews, model experiments and component designs, which are reported in the four individual theses listed in Table 1.3. Where a component result is quoted, the individual thesis is the primary source of that result.

Table 1.3: Individual theses that provide the component-level evidence

Component	Owner (student ID)	Individual thesis focus
SOS Emergency Safety System	De Silva D.S.K (IT22108654)	AI-based tourist emergency detection, nearest-station routing and police response dashboards
AI Itinerary Generator	Wanniarachchi P.M.R (IT22225092)	Random Forest pipeline that turns free-text travel intent into calendar-valid, timed multi-day plans from Sri Lankan attraction and accommodation catalogues
Tourism Risk and Context Intelligence System	Abinaya Rajasekara (IT22090058)	Daily crowd and risk prediction for 197 sites with leakage-aware evaluation on unseen sites
Wellness Component	De Silva D.C.M (IT22108586)	Explainable, multilingual matching of tourists to Ayurveda centres and spiritual retreats

The report is organised as follows. Chapter 2 defines the common problem, its root causes and the four subproblems, and states the group aim and objectives. Chapter 3 presents the integrated solution and its architecture and summarises each component in a common format. Chapter 4 explains how the components are integrated. Chapter 5 describes the shared implementation and how the team executed the project. Chapter 6 reports the group-level evaluation, Chapter 7 the commercialisation, sustainability and responsible-innovation aspects, and Chapter 8 concludes.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

This chapter states the common problem that the team addresses, analyses its root causes and the overall research gap, divides the problem into four non-overlapping subproblems, and sets out the group aim, objectives, scope and constraints.

2.1 Overall problem definition

The common problem is stated as follows, independently of the technology proposed to solve it:

International tourists in Sri Lanka have no single, trustworthy digital service that helps them plan a feasible trip from registered attractions and accommodation, judge how busy a site will be on a given day, choose a suitable and safe wellness provider, and obtain help quickly and with their location and identity when an emergency occurs. As a result, visitors spend hours assembling plans from fragmented sources, crowd at a small number of well-known sites, choose wellness providers without reliable quality or safety information, and lose critical minutes in emergencies explaining who and where they are.

Three pieces of evidence support this statement. First, the published statistics that could support crowd awareness are aggregated: site-level visitor figures are released by SLTDA as annual totals, so no daily, site-level record exists for planning or prediction [1]. Second, the official emergency channels for tourists are voice-based and are not linked to the caller's live location, profile or trip [9], [11]. Third, the wellness sector, Sri Lanka's most distinctive tourism capability for long-stay visitors, lacks standardisation and certification, and visitors cannot easily verify which centres are suitable or safe [13]. Practitioner guides consulted by the itinerary owner describe the planning side of the problem: underestimated mountain-road travel times, fragmented information spread across blogs and messaging applications, and mismatches between hotel prices, meal plans and budgets.

2.2 Root causes and overall research gap

Figure 2.1 summarises the root causes and effects. Five causes were identified.

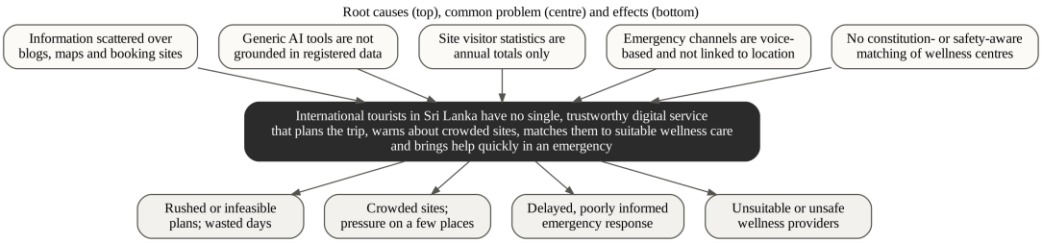


Figure 2.1: Problem tree of the common problem

1. **Fragmented information.** Attractions, opening times, accommodation and prices are spread across blogs, maps and booking sites, and the visitor has to connect them by hand.
2. **Ungrounded automation.** Generic AI chat planners produce fluent itineraries, but they are not grounded in a national register and can suggest venues, times or prices that do not exist.
3. **Missing daily data.** Site-level visitor statistics are annual totals, so neither tourists nor planners can see how busy a site will be on a given day.
4. **Voice-only emergency channels.** Hotlines require the tourist to know the number, speak the operator's language and describe a location they may not know.
5. **No constitution- or safety-aware matching.** Wellness portals list centres but do not consider the visitor's condition and Ayurvedic constitution, and do not summarise quality or safety from guest reviews.

The literature reviewed in the four theses shows that each cause has been studied in isolation. Tourism recommender systems rank items but seldom integrate preference elicitation, recommendation and scheduling [6]; trip-design and orienteering planners produce good routes once interest scores are known but assume those scores are given [14]; crisis informatics classifies emergency messages for situational awareness rather than routing an individual's request [15]; and evaluations of models on repeated measurements are often over-optimistic when random splits are used [16]. The **overall research gap** is therefore the absence of an integrated, evaluated platform for Sri Lanka in which planning, crowd awareness, emergency response and wellness matching share one tourist identity and one data layer, each decision is grounded in registered Sri Lankan data, and each AI component is evaluated honestly

against simple baselines. Closing this gap requires all four components: removing any one leaves the tourist to fall back on a disconnected tool for that part of the trip.

2.3 Subproblem and component mapping

The common problem was divided into four subproblems that do not overlap in what they decide but depend on each other through the tourist identity, the trip and the shared data (Figure 2.2, Table 2.1).

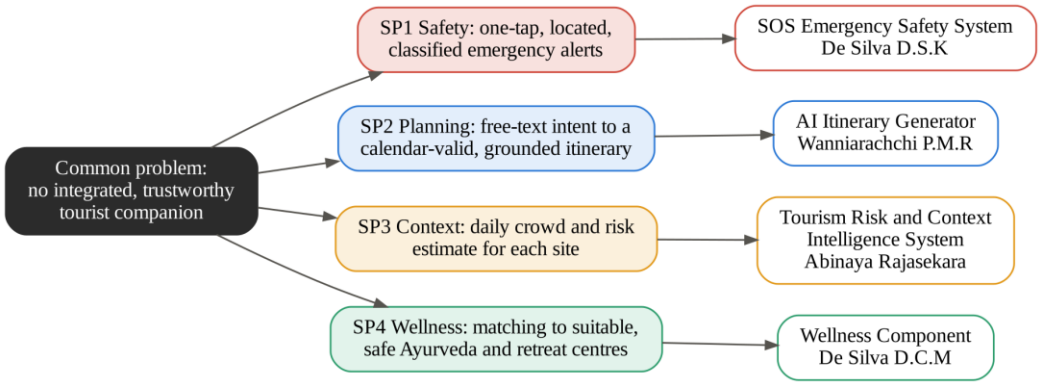


Figure 2.2: Mapping of the common problem to four subproblems and components

Table 2.1: Subproblems, the decision each one makes and the dependencies between them

Subproblem	Decision made	Depends on	Used by
SP1 Safety (SOS)	What kind of emergency has occurred, which responder is needed and which station is nearest	Shared identity and profile; GPS	Police and immigration dashboards
SP2 Planning (Itinerary)	Which registered attractions, cities, nights, visit slots and hotels form a feasible plan	Shared identity; catalogues	Risk itinerary check; SOS trip context
SP3 Context (Risk)	How busy a site is likely to be on a date, and its Low/Medium/High risk class	Site metadata; calendar	Itinerary plans; SOS tourist client
SP4 Wellness (Wellness)	Which Ayurveda centre or retreat best fits the tourist's condition, constitution and budget	Shared identity; centre data	Itinerary trip (stay duration, future)

All four subproblems are necessary. A plan without crowd context sends visitors to the same crowded places at the same time; crowd advice without a plan is difficult to act on; a wellness match without a plan leaves a multi-week stay outside the itinerary; and none of these services is acceptable to a visitor, or to the authorities,

without a dependable way to raise an emergency. Conversely, the subproblems do not overlap: no component makes a decision that another component also makes, which allowed each student to research and evaluate one component independently.

2.4 Overall aim and group objectives

Aim. To design, implement and evaluate an integrated, AI-driven tourism platform for Sri Lanka in which itinerary planning, crowd and risk prediction, emergency response and wellness matching share one tourist identity and one data layer, and in which every AI decision is grounded in registered Sri Lankan data and evaluated against simple baselines.

The aim is achieved through the seven group objectives in Table 2.2. Each objective is mapped to the components that deliver it and to the evidence used to evaluate it in Chapter 6.

Table 2.2: Group objectives, delivering components and evaluation evidence

ID	Group objective (measurable)	Components	Evaluation evidence
GO1	Deliver GPS-anchored, classified emergency alerts to the nearest suitable station in real time, never losing an alert when AI or the network fails	SOS	Hold-out metrics against baselines; fallback and offline tests
GO2	Generate calendar-valid, catalogue-grounded multi-day itineraries with no overlapping visits	Itinerary	Cross-validated metrics; scenario checks; test cases TC1–TC8
GO3	Predict a daily crowd score and risk class for sites, with accuracy reported on unseen sites	Risk	Site-holdout, site-grouped and temporal validation against baselines
GO4	Match tourists to suitable and safe wellness centres with explained scores in five languages	Wellness	Model comparison, ablation study, automated tests
GO5	Operate the four components as one platform: one login, one back end, one database, one entry point	All	Interface and integration tests (Section 6.2)
GO6	Make every model reproducible and evaluate it against simple baselines with the evaluation design stated	All	Retraining scripts; baseline tables (Section 6.3)
GO7	Validate the complete platform with representative users	All	Integrated UAT with SUS (Section 6.4)

2.5 Scope assumptions and constraints

In scope. On-shore leisure travel by international tourists in Sri Lanka; registered attractions and accommodation in the itinerary catalogues (343 attractions in 26 destinations and 921 accommodation units); 197 tourist sites in the crowd model; 83 curated Ayurveda and meditation centres; emergency alerts routed to seeded police, medical and fire stations; a web application that runs in a desktop or mobile browser.

Out of scope. Visa processing, flight booking and payments; live booking of hotels or wellness programmes; connection to the 119 and 1912 call centres or to police operational systems; integration with immigration databases; live traffic, weather or monsoon routing; medical diagnosis; and native mobile applications.

Assumptions. The curated catalogues and site lists are accurate at the time of training; tourists provide truthful dates, budgets and health information; tourists grant location permission when they raise an SOS; and the shared server has the Python and database versions that the components were trained with.

Constraints. Four significant data constraints shape the results and are reported openly in Chapter 6: the itinerary training trips and the SOS statement corpus are synthetic; the crowd target is a constructed index rather than observed visitor counts; and the dosha training profiles are expert-rule variations of six real cases. The project was further constrained by an academic timeline, no external funding, and ethical limits on using real emergency statements or real patient data. Operationally, the components depend on one another through the shared login, database and back end, so a configuration error in the shared environment (for example a library version mismatch) affects every component.

3 OVERALL SOLUTION STRUCTURE

This chapter presents Tour Ceylon as one system. It describes the integrated solution and its value proposition, the high-level architecture, a summary of each component in a common format, and the end-to-end flow of a user and of the data through the platform.

3.1 Integrated solution overview

Tour Ceylon is a web platform that accompanies an international tourist through the whole trip. After registering once, the tourist describes the trip in their own words and receives a timed, day-by-day itinerary built from registered attractions and accommodation; can check how busy a site is likely to be on a given date and receive a risk advisory; can be matched to a suitable Ayurveda centre or meditation retreat for a condition, constitution and budget; and, if something goes wrong, can raise a GPS-anchored SOS that is classified and delivered to the nearest police station's dashboard.

The central value proposition of the integrated platform is **one trusted companion, grounded in Sri Lankan data, for planning, staying aware, staying well and getting help**. Three design principles are shared by all four components and make the platform more than four separate products:

1. **Grounding.** Every recommendation refers to a registered entity in a curated dataset (an attraction, hotel, site, station or wellness centre); no component generates venues, times or prices.
2. **One identity and one data layer.** The tourist logs in once; trips, incidents and wellness sessions are stored against the same identity in one PostgreSQL database, so information entered once is available to the other components.
3. **Fail-safe AI.** Each component keeps working when its model is unavailable or uncertain: the SOS alert is delivered with a keyword fallback label, the itinerary service refuses with a clear HTTP 503 rather than showing a non-ML plan as ML output, the crowd service returns "not covered" for unknown destinations, and the wellness service falls back to its JSON data when the database is unreachable.

A fourth, methodological principle was also shared: each student followed Design Science Research (DSR) [17], [18], building an artefact and evaluating it against simple baselines, and each component uses Random Forest models [19] as its primary learner, which suits medium-sized tabular and sparse text data [20] and gave the team a common vocabulary for model design, evaluation and maintenance.

3.2 High level architecture

Figure 3.1 shows the high-level architecture. It has three layers and a set of external services.

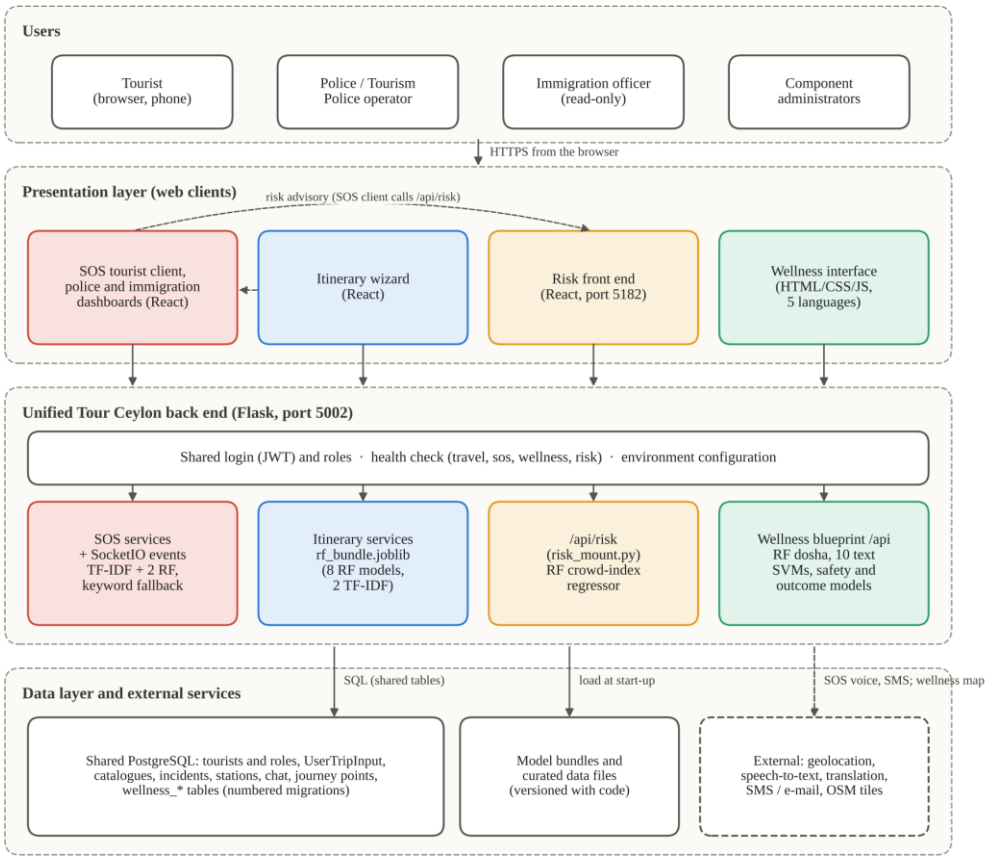


Figure 3.1: High-level architecture of the integrated Tour Ceylon platform (short dashed arrow: token exchange from the itinerary client to SOS)

- **Presentation layer.** Web clients for each component: the itinerary wizard (React), the SOS tourist client with the police and immigration dashboards (React), the risk front end (React, development port 5182) and the wellness interface (HTML, CSS and JavaScript in five languages). The platform home

page links to each module, and the itinerary client opens the SOS module through a token exchange so that the tourist does not log in twice.

- **Application and intelligence layer.** One unified Flask application on port 5002 registers the routes of all four components and exposes a common health check that reports the travel, SOS, wellness and risk components. Each component keeps its models in memory after loading them once at start-up: the itinerary bundle `rf_bundle.joblib` (eight Random Forest models and two TF-IDF vectorisers), the SOS bundle `sos_models.joblib` (TF-IDF with two Random Forests), the risk Random Forest regressor behind `/api/risk`, and the wellness dosha classifier, ten per-language text classifiers, safety classifier and outcome regressor. Real-time SOS events use Flask-SocketIO.
- **Data layer.** A shared PostgreSQL database holds tourists and roles, trip inputs, the attraction and accommodation catalogues, SOS incidents, stations, chat and journey points, and the four wellness tables. Model bundles and curated data files (Excel catalogues, the 197-site dataset, the centre and review files) are stored with the code.
- **External services.** Browser geolocation, speech-to-text and translation for the SOS voice path, SMS and e-mail notifications, and OpenStreetMap tiles for the wellness map.

The boundaries of deployment are the browser on the tourist's or officer's device, the application server that hosts the unified back end and the front-end builds, and the database server. Each component can also run as a standalone server for development and demonstration (for example the wellness component on port 5003) using the same JSON contract as the unified application.

3.3 Component 1 summary

Table 3.1: Summary of Component 1, SOS Emergency Safety System

Aspect	Summary
Owner	De Silva D.S.K (IT22108654)
Subproblem	SP1 Safety: getting the right help to a tourist quickly, with location, identity and a structured description of the incident

Aspect	Summary
Input	One-tap SOS by an authenticated tourist, a typed or spoken statement (translated to English when needed) and the browser GPS position
Data	A corpus of 264 synthetic, labelled statements covering 14 incident types and 3 responder channels (police, medical, fire); seeded police and medical station coordinates
Core method	Statement cleaning, TF-IDF (unigrams and bigrams, 4,000 features) and two Random Forests (180 trees, balanced class weights) for incident type and responder; keyword-rule fallback; haversine nearest-station routing; Flask-SocketIO dashboards; offline queue with duplicate-safe replay; JWT and role-based access
Output	An incident on the nearest station's police dashboard with type, confidence, urgency, cleaned statement, tourist profile, map link and optional Safe Journey route history; status updates and chat with the tourist; a read-only immigration view of consented events
Key result	Responder Random Forest accuracy about 0.811 and macro-F1 about 0.684 (dummy 0.265); incident-type Random Forest macro-F1 about 0.359, below keyword rules (0.496) and Logistic Regression (0.707)
Contribution to the whole	Makes the platform safe to use: every tourist identity created for planning or wellness can raise an alert that reaches responders with the information they need

3.4 Component 2 summary

Table 3.2: Summary of Component 2, AI Itinerary Generator

Aspect	Summary
Owner	Wanniarachchi P.M.R (IT22225092)
Subproblem	SP2 Planning: turning a traveller's free-text wishes and constraints into a feasible, timed multi-day plan grounded in registered inventory
Input	Arrival and departure date and time, free-text experience description, nightly budget in LKR, comfort level and meal plan (BB, HB or FB)
Data	attractions.xlsx (343 attractions, 1,096 timed visit rows, 26 destinations); accomadation.xlsx (921 units, 890 usable, with BB/HB/FB price bands); a synthetic training workbook of 171 trips and more than 25,000 labelled rows
Core method	A bundle of Random Forest models (200 trees each) over TF-IDF features: need model (10 moods, 6 categories), attraction relevance, destination selection, nights and order, places per day, visit-slot choice and hotel choice; a deterministic greedy scheduler that guarantees non-overlapping visits and the calendar rule travel_days = total_days − 2
Output	Itinerary rows: airport arrival, hotel check-in with meal-plan price band and stay total, timed attraction visits on travel days, and a departure day for shopping and the flight; a personal saved directory of shops, agencies and guides

Aspect	Summary
Key result	Need model micro-F1 0.793 (5-fold out-of-fold); attraction model ROC-AUC 1.000 and hotel model 0.993 (GroupKFold by trip), interpreted as reproducing the synthetic labelling logic; eight functional test cases passed
Contribution to the whole	Provides the trip (dates, cities, selected places) that gives context to crowd advisories and wellness stays, and the logged-in entry point from which the SOS module is opened

3.5 Component 3 summary

Table 3.3: Summary of Component 3, Tourism Risk and Context Intelligence System

Aspect	Summary
Owner	Abinaya Rajasekara (IT22090058)
Subproblem	SP3 Context: estimating how busy a site will be on a given day when no daily, site-level record exists
Input	Site and date (and, for the itinerary check, a list of planned site visits)
Data	197 sites and 144,007 daily records for 2024–2025 in 28 columns; a constructed daily crowd index as the target; eleven data defects identified and disclosed
Core method	Random Forest regressor (50 trees, depth 12) on 16 features; site-holdout, site-grouped cross-validation and temporal evaluation against mean, category-month mean and Ridge baselines; ablation and shuffled-target checks
Output	A crowd score between 0 and 1 and a Low, Medium or High risk class through `/api/risk`; an itinerary-level risk check; a ranking of 114 green sites; a "not covered" response outside the 197 sites
Key result	On ten unseen sites MAE 0.1232 and R^2 0.3331, against 0.2057 MAE for the best baseline; site-grouped CV over 197 sites R^2 0.5071; random-split R^2 0.8492 shown to be optimistic
Contribution to the whole	Adds a day-level context signal to planning and to the tourist client, and supports the spreading of visitors across sites

3.6 Component 4 summary

Table 3.4: Summary of Component 4, Wellness Component

Aspect	Summary
Owner	De Silva D.C.M (IT22108586)
Subproblem	SP4 Wellness: choosing a suitable, safe Ayurveda centre or meditation retreat for a multi-week stay before travelling

Aspect	Summary
Input	Health condition or retreat goal (selected or described in free text in English, German, Russian, Chinese or Japanese), age, gender, four lifestyle answers, budget tier
Data	83 real Sri Lankan centres with verification notes and 256 guest reviews; 102 tourist profiles derived from six real expert cases; 71 labelled reviews for safety; 54 cases for outcome estimation
Core method	Random Forest dosha classifier selected from six algorithms; transparent weighted matching (condition 35%, quality 25%, dosha 20%, budget 20%); ten TF-IDF and linear SVM classifiers for free text; hybrid keyword and SVM review-safety screening; Random Forest outcome regressor
Output	Ranked centres with a score breakdown, predicted dosha, expected treatment duration and cost, outcome estimate and safety flags, in five languages; an administration dashboard
Key result	Dosha classifier accuracy 0.810, weighted F1 0.794, 5-fold CV 84.3% ± 9.8%; full model raises top-match quality from 3.41 to 4.61 and budget alignment from 50.0% to 90.0% over condition-only matching; 75 of 75 automated test executions pass; median latency 22 ms
Contribution to the whole	Extends the platform to Sri Lanka's highest-value, longest-staying visitor segment, with matching sessions stored against the shared tourist identity

Table 3.2 to Table 3.4 show a deliberate division of work: each component owns one decision, one dataset and one family of models, and all four share the same identity, database and back end. Table 3.5 compares the four components side by side.

Table 3.5: Side-by-side comparison of the four components

Aspect	SOS	Itinerary	Risk	Wellness
Learning task	Text classification (2 tasks)	Multi-model ranking and scheduling	Regression and 3-class risk	Classification, ranking, text classification
Primary model	2 Random Forests + TF-IDF	8 Random Forests + TF-IDF	Random Forest regressor	Random Forest + linear SVMs
Training data	Synthetic statements	Synthetic trips from catalogues	Constructed crowd index	Expert-derived profiles; real centres and reviews
Main baseline	Keyword rules; dummy	Keyword and even-split rules	Mean, category-month, Ridge	Condition-only matching; 6 algorithms
Fail-safe	Keyword fallback; offline queue	HTTP 503 if bundle missing	"Not covered" response	JSON fallback if DB down

Aspect	SOS	Itinerary	Risk	Wellness
Retraining	RETRAIN_SOS_ML.bat	RETRAIN_ALL_RF.bat	validate_model.py and training script	Training notebook

3.7 End to end user and data flow

Figure 3.2 traces a tourist and their data through the complete platform. The numbered steps below follow the persona Anna from Section 1.2.

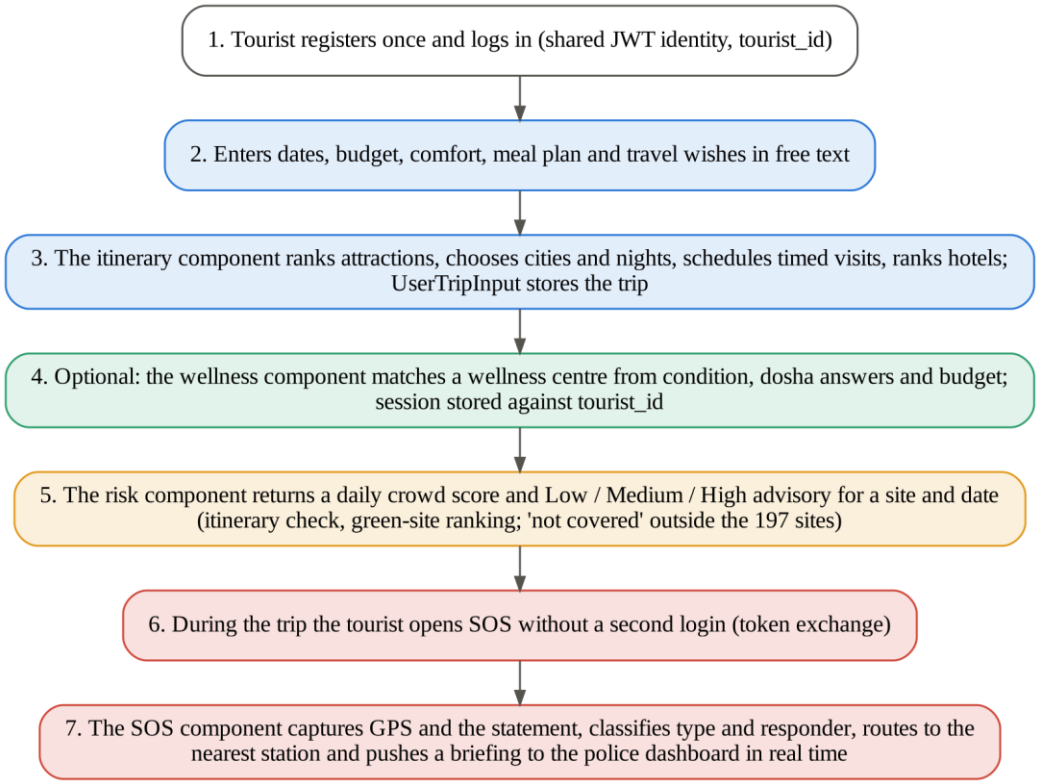


Figure 3.2: End-to-end user and data flow through the four components

1. Anna registers and logs in once. The shared authentication service issues a JSON Web Token (JWT) [21] that carries her identity and role.
2. In the itinerary wizard she enters her arrival and departure, a nightly budget, a comfort level, Half Board and the sentence "beaches, culture and nature". The trip input is stored in the `UserTripInput` table.
3. Itinerary infers her moods and categories, ranks the registered attractions, chooses destinations, nights and order, schedules non-overlapping visits on the

travel days and ranks hotels for each destination with Half Board prices. The itinerary is presented with arrival, check-ins, timed visits and a departure day.

4. From the home page she opens the wellness module. The wellness component asks four lifestyle questions, predicts her dominant dosha, and ranks centres for back pain within her budget tier, with safety flags and an expected duration. The matching session is stored against her tourist identity.
5. For a site she plans to visit, the risk component returns the predicted crowd score and a Low, Medium or High advisory for the date; for a site outside the 197 modelled sites it returns "not covered" rather than a guess.
6. During the trip she opens the SOS module from the itinerary client. The token exchange opens it without a second login, and her profile is already known.
7. If she raises an SOS, the SOS component captures her GPS position and statement, classifies the incident type and responder, finds the nearest suitable station, stores the incident and pushes a structured briefing to the police dashboard in real time. If she is offline, the alert is queued and replayed when connectivity returns.

The components interact at four points: the shared identity (all components), the token exchange from the itinerary client to SOS, the risk advisory requested by the SOS tourist client, and the shared database tables through which the trip and wellness sessions can be read by other components (itinerary and wellness to the platform). Section 4.2 documents these interfaces.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

This chapter explains how the four components were integrated into one platform: the integration strategy, the interfaces and data contracts, the non-functional requirements shared by all components, the integrated workflow with its exception paths, and the contribution and dependency matrix of the team.

4.1 Integration strategy

The team followed a **standalone-first, shared-contract** strategy. Each component was first developed and evaluated as a standalone module with its own REST API, so that each student's research could progress and be examined independently. From the start, however, the components agreed on four shared elements: Python and Flask for the back end, PostgreSQL for persistent data, JSON over REST for every interface [22], and a single tourist identity issued by the shared login. Integration then consisted of mounting each component's routes in the unified Flask application without changing its JSON contract, adding each component's tables to the shared database through numbered migrations, and linking the front ends from the platform home page.

Integration proceeded in the sequence shown in Table 4.1. Interface ownership stayed with the component owner: each student was responsible for keeping their routes, tables and model files compatible with the unified application, while the shared application shell, authentication and configuration were group-owned.

Table 4.1: Integration sequence and the issues found at each step

Step	Integration work	Issue found and resolution
1	Shared login and tourist identity; itinerary trip input in `UserTripInput`	Mismatched trip-input field names between the form and the itinerary builder; resolved by aligning the model
2	Catalogues seeded into PostgreSQL; meal-plan price columns (migration 008)	Destination spelling variants split cities; resolved with a Destination Alias sheet
3	SOS module opened from the itinerary client through a token exchange	None reported; handoff succeeded in bootstrap tests
4	Risk routes mounted under `/api/risk` in the unified back end	Mount file missing; resolved by creating `risk_mount.py`; SOS tourist client confirmed calling the risk endpoints

Step	Integration work	Issue found and resolution
5	Wellness blueprint under `/api`; tables created by migration 005; home-page tile	Shared back end pins scikit-learn 1.5.1 while wellness models are saved with 1.8.0 (and the risk model was trained with 1.7.1); fix communicated; open at submission

Integration risks were managed in three ways. First, each component kept a fail-safe behaviour (Section 4.4), so that an integration fault degraded one service rather than the whole platform. Second, each component was verified through the same JSON contract in standalone and unified mode, so that a component's own tests remained valid after integration. Third, secrets and environment-specific settings were moved into environment variables so that the same code could run in each member's environment and on a shared server.

4.2 Interfaces data contracts and shared models

All interfaces are REST endpoints that accept and return JSON, except the real-time SOS events, which use Flask-SocketIO. Table 4.2 lists the principal interfaces of the integrated platform; the complete specification is in Appendix A.

Table 4.2: Principal interfaces of the integrated platform

Interface	Provider	Consumer	Contract (summary)
Shared login and token exchange (`POST /api/tourists/token-exchange`)	Platform / SOS	Itinerary client, SOS client	Access token in, SOS session out; no second login
Itinerary suggestion, hotel ranking and generation	Itinerary	Itinerary wizard	Trip input in; ranked attractions, ranked hotels with meal-plan bands, ordered itinerary rows out; HTTP 503 if the bundle is missing
SOS incidents (`POST /api/incidents`, police lifecycle actions, chat, live location)	SOS	SOS tourist client, police dashboard	GPS, statement, optional journey points in; incident with type, confidence, urgency, station out
SocketIO events (`incident:new`, `incident:status`, `incident:location`, `chat:message`)	SOS	Police dashboard, tourist client	Event name and JSON payload per station room

Interface	Provider	Consumer	Contract (summary)
Crowd and risk (`/api/risk`, `/api/risk/itinerary-check`, `/api/risk/green-sites`)	Risk	SOS tourist client, risk front end; itinerary (planned)	Site and date in; crowd score 0–1 and risk class out; "not covered" for unknown sites
Wellness (`POST /api/recommend`, `/api/match`, `/api/classify-symptom`, `/api/safety-check`, `/api/cost-estimate`)	Wellness	Wellness interface; platform	Condition, answers and budget in; ranked centres with score breakdown, dosha, duration and flags out
Health check	Platform	Operators	Status of the travel, SOS, wellness and risk components

Shared data model and identifiers

Figure 4.1 shows the shared data model at the level of the entities that cross component boundaries. The **tourist identity** is the key shared identifier: it owns trip inputs, raises incidents and links wellness matching sessions (`wellness_matching_sessions.tourist_id`). The **trip** (`UserTripInput`) carries the trip identifier, dates and selected places, which other components can read. The **site** is the key of the risk model; the itinerary catalogue and the 197-site model use different site lists, and 142 of 350 catalogued destinations are outside the risk model, which is why the risk service returns an explicit "not covered" response.

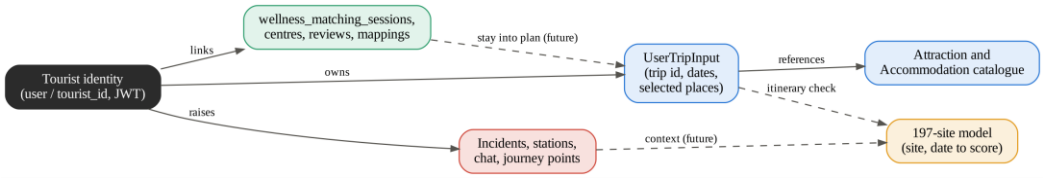


Figure 4.1: Entities and identifiers shared across components (dashed lines are planned or partial links)

Validation and error-handling responsibilities follow the provider: each provider validates its own inputs and returns an explicit error. The wellness API validates condition, answers and budget before scoring; the SOS API rejects an alert without GPS and returns guidance with the 119 and 1912 numbers; the risk API returns "not covered" for unknown sites; and the itinerary API returns HTTP 503 when its model bundle is missing. Versions are controlled through the shared repository, numbered

database migrations (for example 005 for wellness and 008 for meal-plan prices) and versioned model files.

4.3 Shared nonfunctional requirements

The non-functional requirements of the complete platform were organised by the ISO/IEC 25010 quality characteristics [23]. Table 4.3 records each shared requirement, how the components meet it and the evidence available.

Table 4.3: Shared non-functional requirements of the integrated platform

Characteristic	Shared requirement	How it is met	Evidence status
Performance	Interactive responses for planning and matching; real-time delivery of SOS alerts	Models loaded once and kept in memory; light Random Forest and linear models	Wellness median 22 ms; itinerary target under 30 s; risk green-site ranking about 7 s; SOS latency not yet measured
Reliability	No AI failure may block an alert or show an incorrect plan	Keyword fallback and offline queue (SOS); HTTP 503 (itinerary); "not covered" (risk); JSON fallback (wellness)	Verified by component tests
Security	Only authenticated users reach their own data; secrets outside code	JWT [21] and role-based access; environment variables; admin passwords configured centrally; OWASP Top 10 as a reference [24]	Authorisation-matrix test (SOS); input-validation tests (wellness); no penetration test
Privacy	Minimum personal data, consent-based sharing, compliance with the PDPA [25]	Location shared only at SOS time; consented immigration view; synthetic training data; no personal data in crowd data	Design level; no data-protection impact assessment yet
Usability	Usable on a phone by a tourist under time pressure, in more than one language	Large SOS control; wizard steps; five-language wellness interface; plain-language explanations	Stakeholder and peer review; integrated UAT pending
Maintainability	Models retrained without code changes when data change	One-click retraining scripts and notebooks per component	Exact reproduction of reported metrics confirmed for Risk and Wellness
Compatibility	Components interoperate through one contract	JSON over REST; shared tables and migrations	Library version pin open (Section 6.6)

Characteristic	Shared requirement	How it is met	Evidence status
Scalability	Stateless request handling	Stateless model inference; SocketIO can scale with a message queue	Not load-tested

4.4 Integrated workflow and exception handling

In normal operation the platform follows the flow of Section 3.7. Table 4.4 lists the important failure paths of the integrated workflow and the behaviour designed for each. The common rule is that a failure in an AI component degrades the quality of a single answer but never stops the tourist from completing the task or reaching help.

Table 4.4: Failure paths of the integrated workflow and designed recovery behaviour

Failure	Where	Designed behaviour
Model bundle cannot be loaded	SOS	Keyword classifier labels the incident; alert still created, routed and notified
Model bundle missing	Itinerary	Itinerary endpoints return HTTP 503 with a clear message; no rule-based plan is shown as ML output
No network when SOS is raised	SOS	Alert stored in the browser and replayed on reconnection; server-side duplicate check prevents double delivery
GPS permission denied	SOS	SOS blocked with guidance on enabling location; 119 and 1912 shown on screen
Speech-to-text fails	SOS	Audio preserved and attached to the incident; incident created with GPS and a fallback label
Site not in the crowd model	Risk	"Not covered" response; no prediction is invented
Database unreachable	Wellness	Centres loaded from the JSON file; recommendations continue; warning logged
Free-text symptom has no recognisable signal	Wellness	No-signal flag returned; tourist asked to choose a condition
Library version mismatch in shared environment	All	Model fails to load at start-up; component reports unhealthy in the health check (fix: pin versions; Section 6.6)
Duplicate SOS within a short interval	SOS	Second alert ignored or merged

4.5 Team contribution and dependency matrix

Table 4.5 maps each student to their component, research tasks, development tasks, integration work and test evidence. Table 4.6 shows the dependencies between components. The agreed contribution declaration is in Appendix B.

Table 4.5: Team contribution matrix

Member	Research	Development	Integration work	Test evidence
De Silva D.S.K (SOS)	Emergency-text classification; tourist safety; data protection	SOS client, police and immigration dashboards, routing, SocketIO, offline queue	Token exchange from the itinerary client; shared tourist identity in briefings	Hold-out metrics; automated SOS tests; scenarios
Wanniarachchi P.M.R (itinerary)	Itinerary literature; synthetic-label design; RF pipeline evaluation	Training and runtime package, Flask services, attraction, accommodation and itinerary pages, saved directory	Trip input model, catalogue seeding, migration 008, alias sheet	TC1–TC8; scenario checks; training log
Abinaya Rajasekara (risk)	Crowd prediction; grouped validation; data audit	197-site dataset, crowd index, RF model, `/api/risk` service, React front end	`risk_mount.py`; risk routes called by the SOS client	validate_model.py checks; site-holdout and grouped CV
De Silva D.C.M (wellness)	Wellness tourism; dosha classification; review-based quality and safety	Centre dataset, models, matching API, five-language interface, admin dashboard	Wellness blueprint, migration 005, home-page tile, version-pin report	48 test cases (75/75); latency benchmark; ablation
Shared	Common problem, architecture and this report	Application shell, authentication, home page	Component monorepo and itinerary integration branch created by Wanniarachchi P.M.R; unified back end on port 5002; each owner mounted and tested their own component	Integrated E2E tests and UAT (Section 6.2)

Table 4.6: Dependency matrix between components (row depends on column)

	SOS	Itinerary	Risk	Wellness	Shared platform

	SOS	Itinerary	Risk	Wellness	Shared platform
SOS	–	Entry point and token	Risk advisory in client	None	Login, DB, back end
Itinerary	Opens SOS	–	Itinerary check (planned)	Stay duration (planned)	Login, DB, back end
Risk	None	Planned visits	–	None	Back end
Wellness	None	None	None	–	Login, DB, back end, home page

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

This chapter describes the technology and infrastructure that the four components share, the development and integration process, how the integrated solution is configured and deployed, and the evidence of how the team executed the project.

5.1 Common technology stack and infrastructure

Table 5.1 lists the common technology stack and the reason for each major choice. The choices were made early so that four separately developed components could be combined without rewriting.

Table 5.1: Common technology stack and justification

Layer	Technology	Reason for the common choice
Language	Python 3.10 or later; JavaScript	One language for the machine-learning code and the back end, so models run in-process without a separate inference service
Back end	Flask, Flask blueprints and mounts, SQLAlchemy / psycopg2, Flask-SocketIO	Lightweight framework that lets each component register its routes in one application; real-time events for SOS
Machine learning	scikit-learn [26], pandas, NumPy, joblib	Random Forests, TF-IDF, SVMs and baselines in one well-documented library; simple serialisation of model bundles
Database	PostgreSQL	Relational integrity and one store shared by all components; numbered migrations per component
Front end	React with Vite (SOS, itinerary, risk); HTML, CSS and JavaScript (wellness); Leaflet with OpenStreetMap	Component reuse and responsive layouts; the wellness interface kept framework-free for a five-language static client
Authentication	JSON Web Tokens; role-based access	Stateless sessions and a single login across components
Tooling	Git and a shared repository; Visual Studio Code / Cursor; Google Colab and Jupyter; pytest	Version control, notebooks for reproducible training, automated tests

The shared infrastructure consists of the unified Flask application, which registers the routes of the four components and runs on port 5002; the shared PostgreSQL database; the shared repository, which holds the four components side by side; and

environment configuration files (a `.env` file per environment, with a committed template) that hold database connection strings and secrets. The model files of each component are versioned with its code so that the code and model always match.

5.2 Development and integration process

Development followed an iterative, increment-based process within the Design Science Research cycle [18]: each increment added or improved one capability, re-ran the component's tests and evaluation, and was reviewed at supervision meetings and progress presentations. The following practices were common to the team.

- **Interface-first work.** Each component defined its JSON request and response formats before its user interface, which allowed other members to call it from their clients and allowed the unified back end to mount it without changes.
- **Version control.** All code lived in a shared Git repository. Members worked on their own component folders and branches and merged into the shared branch for integration, following the group's shared branch workflow.
- **Separation of training and serving.** No component trains a model on a user request. Training runs offline through a script or notebook (`RETRAIN_SOS_ML.bat` for SOS, `RETRAIN_ALL_RF.bat` for the itinerary, `validate_model.py` and the training script for risk, a Colab notebook for wellness), and the serving code only loads the saved bundle.
- **Feedback-driven change.** Supervisor and panel feedback changed the work in visible ways. For example, the progress-presentation request to show the trained models and their parameters led the wellness owner to build a single reproducible training notebook, and supervisor feedback on evaluation led the itinerary owner to discuss the near-perfect attraction AUC critically.
- **Issue handling.** Integration issues were recorded and fixed by the owner of the interface concerned (Table 4.1); open issues are listed in Section 6.6.
- **Release practice.** A component was considered ready for integration when its tests passed in standalone mode and its routes answered through the unified back end.

Continuous integration with automated builds was not set up; tests were run by each owner before merging. Planning followed the milestones of the module: the first and second progress presentations (PP1 and PP2), supervisor meetings between them, and the final submission. Within those milestones each member planned their own component work, and integration tasks, deadlines and submission instructions were agreed in the group's shared messaging channel. Each component was developed on its own branch (`itinerary`, `sos`, `risk_management` and `wellness` in the R26-IT152 repository; `itinerary_generator`, `sos_system`, `risk_management` and `wellness` in the `tour_ceylon` component monorepo), and merges into the main branch were made through pull requests.

5.3 Deployment and operational configuration

Figure 5.1 shows the deployment view of the integrated solution.

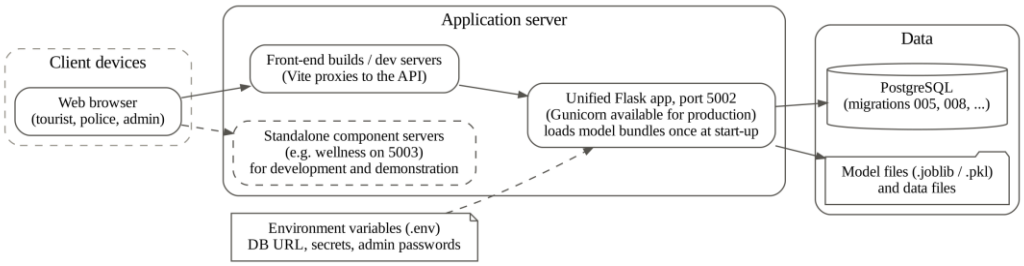


Figure 5.1: Deployment view of the integrated solution

Build and configuration. The back end is configured through environment variables: the database connection, secrets and component administration passwords (for example `WELLNESS_ADMIN_PASSWORD`). The front ends are built with Vite or served as static files, and during development the Vite servers proxy API calls to the unified back end. Database tables are created by numbered migrations, and the catalogues and station data are loaded by seed scripts (`seed_data.py` for the itinerary catalogues and `seed_sos_data.py` for police and medical stations).

Start-up. The unified back end is started once (for example with `RUN_BACKEND.bat`) and loads each component's model bundle into memory. A health check reports the status of the travel, SOS, wellness and risk components. The shared back end includes Gunicorn as a production WSGI server.

Operation and updating. When source data change, the owner updates the data file, re-seeds the database if needed, runs the component's retraining script and restarts the back end. No code change is required for catalogue updates, which was verified for meal-plan prices in the itinerary component.

Monitoring, backup and restore. The prototype relies on server logs and the health check for monitoring. Backup and restore of the PostgreSQL database use standard database dump and restore; model files and curated data are restored from the repository. A production deployment would add TLS, managed PostgreSQL with automated backups, centralised logging and an audit log. At the time of writing the integrated platform is run and demonstrated on the members' own Windows machines, with the unified back end started by `RUN_BACKEND.bat` and a local PostgreSQL database; a hosted deployment on a university or cloud server has not yet been set up and is part of the deployment work in Section 8.3.

5.4 Evidence of team execution

Table 5.2 summarises the evidence of how the team executed the project. Raw activity logs are not reproduced here; they are available for inspection on request.

Table 5.2: Summary of team execution evidence

Evidence source	What it shows	Summary
Progress presentations (PP1, PP2)	Staged review of each component and of the platform	Feedback recorded and acted on, for example the reproducible training pipeline (wellness) and the critical evaluation of optimistic metrics (itinerary, risk)
Supervision meetings	Guidance on evaluation design and honest reporting	Reflected in every individual thesis: baselines, unseen-data evaluation and disclosed limitations
Git history of the shared repository	Commits per member, merges into the shared branch, integration fixes	Contributor history of the shared R26-IT152 repository (Figure 5.3); component branches for SOS, itinerary, risk and wellness; itinerary component merged on 31 August and risk-management pull requests #2 to #4 merged on 1 September 2026; wellness models retrained and scikit-learn 1.8.0 pinned on 9 October 2026

Evidence source	What it shows	Summary
MS Planner / Teams	Task allocation, deadlines and communication	Team R26-IT-152 - Research in Microsoft Teams with an MS Planner plan per component (Figure 5.2); the plan shown holds 70 tasks in seven buckets, prioritised as urgent, important and medium, all completed (0 tasks left)
Research diaries and meeting records	Individual progress and decisions	Each member's progress, design decisions and the integration issues they resolved are recorded in their individual thesis; supervisor feedback from each meeting and progress presentation was acted on as summarised in the first two rows
Project website	Shared public output of the group	Tour Ceylon website (R26-IT-152) with domain, milestones, documents, presentations and team pages (Figure 5.4)
Stakeholder records	External engagement	Domain-expert input (wellness); proposal and letters connected with Police Headquarters and immigration officials as reported by the SOS component owner

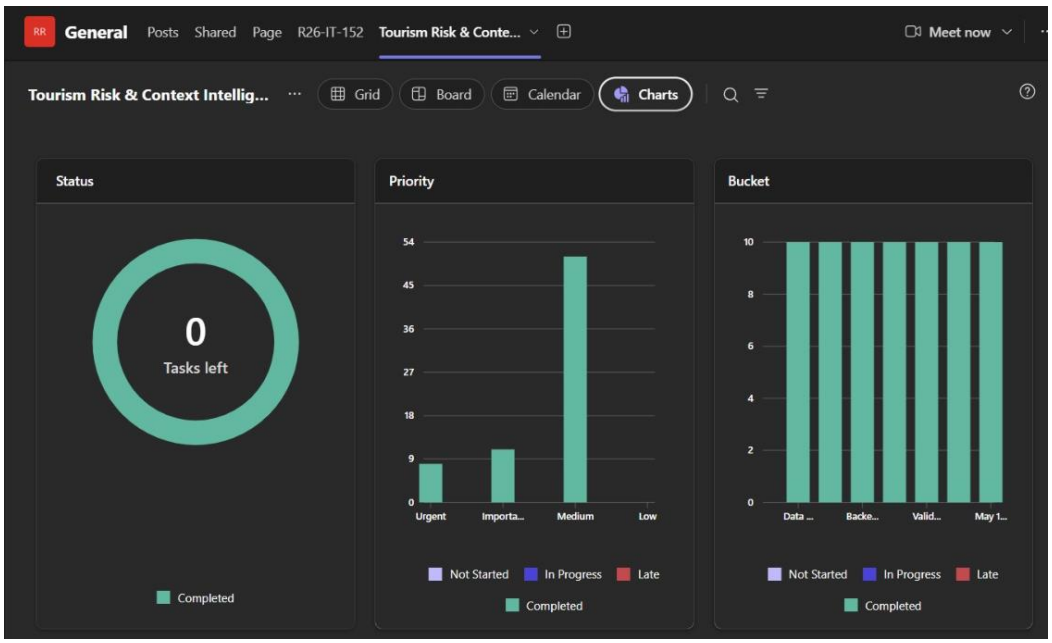


Figure 5.2: MS Planner charts of a component plan in the R26-IT-152 Research team

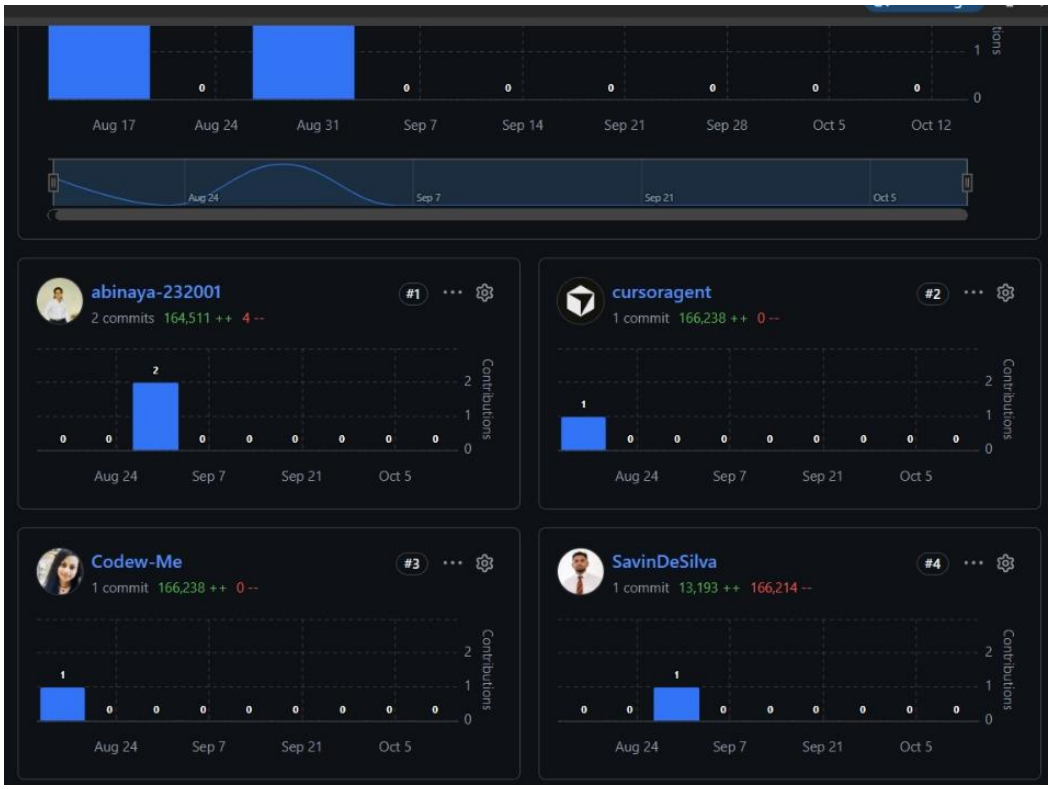


Figure 5.3: Contributors to the shared R26-IT152 GitHub repository

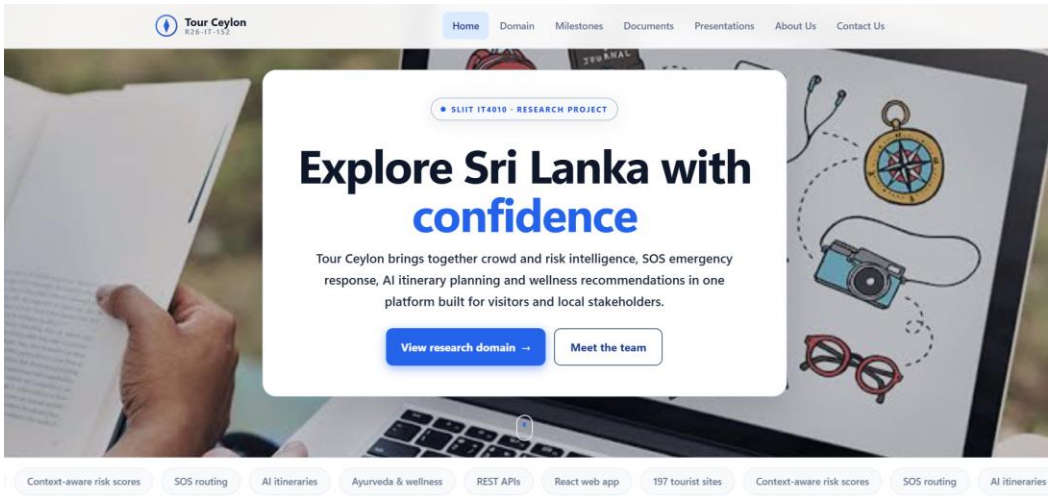


Figure 5.4: Home page of the Tour Ceylon project website

Integration work itself provides objective evidence of collaboration: five integration steps were completed and the issues found were resolved by the owners concerned (Table 4.1), and every component exposes its functions through the unified back end.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

This chapter defines the group-level evaluation framework, reports the integration and end-to-end testing, synthesises the overall results of the four components, reports stakeholder and user validation, maps the evidence to each group objective and discusses the limitations of the integrated solution. Detailed component experiments are reported in the individual theses.

6.1 Evaluation framework and success criteria

Four group-level evaluation questions (EQ) were defined before the results were collected. Table 6.1 lists them with their acceptance criteria, evaluation methods and data.

Table 6.1: Group-level evaluation questions, success criteria and methods

Question	Success criterion	Method and data
EQ1 Does each component's AI beat simple baselines on data it was not trained on?	Primary metric better than the strongest simple baseline under the stated evaluation design	Hold-out, cross-validation or grouped validation in each component, against keyword, mean, dummy or condition-only baselines
EQ2 Do the four components operate as one platform?	One login, all routes served by the unified back end, shared tables created, home page links every module	Interface and integration tests; health check
EQ3 Does the platform remain usable when an AI part or the network fails?	Every failure path in Table 4.4 produces the designed behaviour	Fallback, offline and error-path tests in each component
EQ4 Do representative users find the complete platform usable and useful?	Mean System Usability Scale score of at least 68 [27], [28]; all core tasks completed	Task-based integrated UAT with tourists and responders

The evaluation design of each result is always stated, because the same model can look very different under different designs. Random splits that place records from the same site, trip or statement template in both training and test sets give optimistic estimates when observations are grouped [16], and labels that are generated from the same features that the model uses can make a task appear easier than it is [29]. The

team therefore reports grouped or unseen-data results as the primary evidence and treats optimistic designs as references only.

6.2 Integration and end to end testing

Table 6.2 reports the integration evidence available at the time of writing. Each component has been verified through the same JSON contract that the unified back end exposes, and the integration points between components have been checked individually.

Table 6.2: Integration and interface test evidence

Test	Scope	Result
Wellness API through the shared contract	Wellness, 48 test cases, 75 executions (pytest)	75 of 75 passed; median recommendation latency 22.19 ms
Wellness tables created by migration 005	Wellness and shared database	Passed: four wellness tables created
Itinerary functional tests TC1–TC8	Itinerary through the API and user interface	8 of 8 passed (calendar rules, no overlaps, meal-plan prices, HTTP 503, saved directory)
Trip input consistency across steps	Itinerary and shared database	Passed after field names were aligned
Risk routes in the unified back end	Risk via `risk_mount.py`	Passed after the missing mount was created
SOS tourist client calling the risk service	SOS to risk	Passed: calls confirmed in the browser developer tools
Token exchange from itinerary client to SOS	Itinerary to SOS	Passed in bootstrap tests
SOS automated tests (classification, fallback, duplicates, authorisation matrix, alerts feed)	SOS	Test modules implemented by the owner: `test_sos_ml.py`, `test_priority_one_dispatch_and_duplicates.py`, `test_authorization_matrix.py`, `test_tourist_alerts_feed.py`
Risk model verification	Risk: reproduction, shuffled target, read-only check	Passed: locked result reproduced exactly; no pipeline leak
Wellness models in the unified environment	Wellness with shared scikit-learn version	Open: requires pinning scikit-learn 1.8.0 in the shared requirements

An end-to-end test across all four components in the unified deployment is defined in Table 6.3. The scenarios follow the flow of Section 3.7 and the failure paths of Section 4.4. The scenarios are executed on the final integrated deployment before the final presentation.

Table 6.3: Integrated end-to-end test scenarios

ID	Scenario	Expected result	Result
E2E-1	Register, log in, generate a 7-day itinerary with Half Board	Valid itinerary; no visits on arrival or departure day; HB prices	Planned
E2E-2	From the same session, request a wellness match for back pain	Ranked centres with score breakdown; session stored with the tourist ID	Planned
E2E-3	Check the crowd advisory for a planned site and for a site outside the model	Score and class for the first; "not covered" for the second	Planned
E2E-4	Open SOS from the itinerary client and raise a text alert	No second login; incident on the nearest station dashboard with profile and map link	Planned
E2E-5	Raise an SOS while offline, then reconnect	Alert replayed once; no duplicate	Planned
E2E-6	Stop the database and request a wellness match	Results from the JSON fallback; warning logged	Planned
E2E-7	Remove the itinerary bundle and request an itinerary	HTTP 503 with a clear message; other components unaffected	Planned
E2E-8	Health check after start-up	All four components reported healthy	Planned

Security testing at platform level is limited to the role-based authorisation tests of the SOS component and the input-validation tests of Wellness. No penetration test, load test or regression suite across all components was carried out; these are listed as future work.

6.3 Overall results

Table 6.4 synthesises the principal result of each component under its primary evaluation design, together with the strongest simple baseline. Figure 6.1 shows the same comparison graphically. The metrics differ between components because the

tasks differ; the common question is whether each AI component improves on what a simple rule or average would achieve.

Table 6.4: Principal results of the four components against simple baselines

Component and task	Evaluation design	AI model result	Strongest simple baseline	EQ1 outcome
SOS Responder routing (3 classes)	Stratified hold-out (~53 statements)	Accuracy ≈0.811; macro-F1 ≈0.684	Dummy macro-F1 ≈0.265	Met
SOS Incident type (14 classes)	Stratified hold-out	RF macro-F1 ≈0.359; LR ≈0.707	Keyword rules ≈0.496	Not met by served RF; met by LR
Itinerary Need detection from text	5-fold out-of-fold	Micro-F1 0.793	Keyword suggestion (qualitative)	Not measured numericall y
Itinerary Attraction and hotel ranking	GroupKFold by trip	AUC 1.000 and 0.993	Keyword / highest rating (qualitative)	Optimistic: labels share features
Risk Crowd score, unseen sites	10-site holdout	MAE 0.1232; R^2 0.3331	Ridge MAE 0.2057; category-month R^2 0.1201	Met
Risk Crowd score, all sites	Site-grouped CV (5×3)	MAE 0.1159; R^2 0.5071	Ridge MAE 0.1452; R^2 0.3883	Met
Wellness Dosha classification	Stratified hold-out and 5-fold CV	Accuracy 0.810; F1 0.794; CV 84.3% ± 9.8%	Five other algorithms (lower)	Met
Wellness Matching quality	Ablation, 60 queries	Top-match quality 4.61; budget fit 90.0%; dosha fit 51.7%	Condition only: 3.41; 50.0%; 10.0%	Met
Wellness Review safety screening	Leave-one-out, 71 reviews	Precision 1.0; recall 0.077	Keyword rules (higher recall)	Used as second opinion only

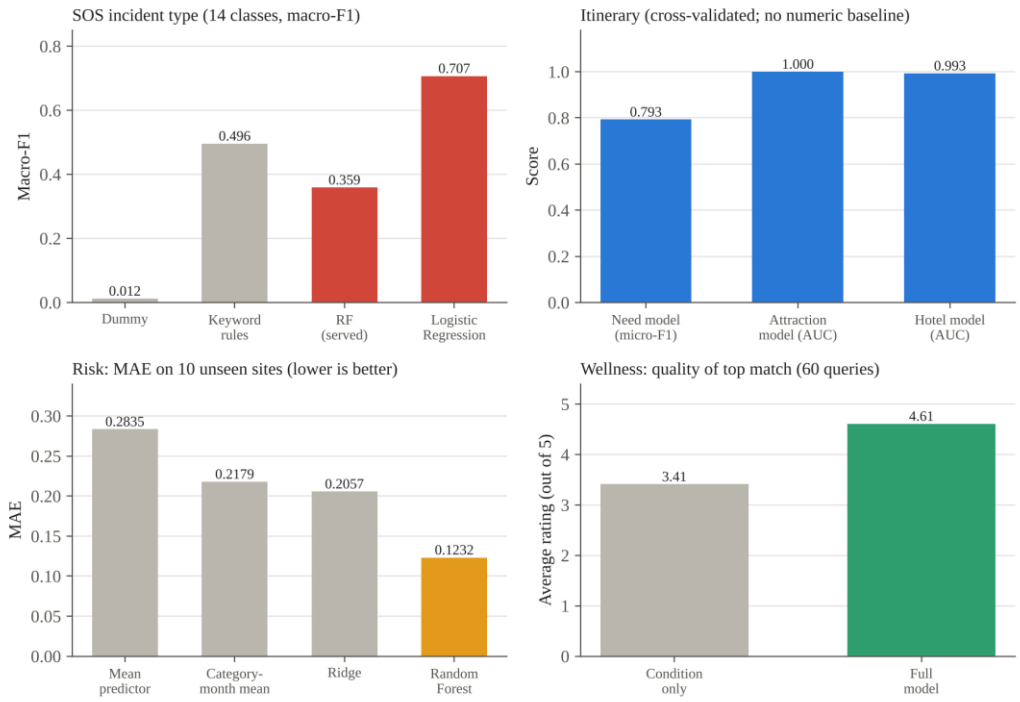


Figure 6.1: Principal result of each component under its primary evaluation design (grey bars are baselines)

Three findings emerge at group level. First, **the AI components improve on simple baselines where the evaluation is strict**: on unseen sites, the risk model reduces the error of the best simple baseline by about 40%; the wellness component's full matching model raises the average quality of the top match from 3.41 to 4.61 out of 5 and budget alignment from 50.0% to 90.0%; and the SOS responder model routes about four in five held-out statements to the correct service, far above a majority-class baseline.

Second, **the honest evaluation designs changed conclusions**. Figure 6.2 shows that the same risk model scores R^2 0.8492 on a random split but 0.3331 on ten unseen sites, because the random split rewards remembering sites already seen. In the SOS component, comparing against keyword rules showed that the served Random Forest is weaker than both keyword rules and Logistic Regression on the fourteen-class task, which led to a clear recommendation to serve Logistic Regression. In the itinerary component, the near-perfect attraction AUC is reported as evidence that the model learned the labelling rule, not as evidence of real-world ranking quality.

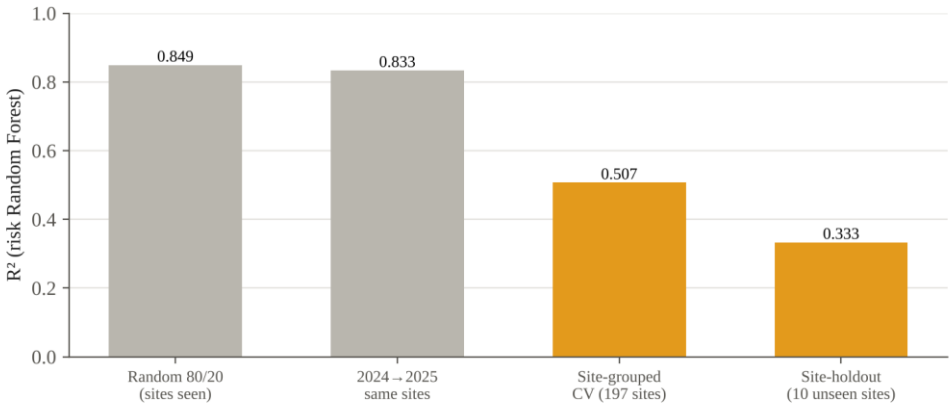


Figure 6.2: R^2 of the risk model under evaluation designs that do and do not include the test sites in training

Third, **the platform-level properties hold**: every component is reachable through the unified back end, all four share the tourist identity and database, and each has a designed fail-safe. Product-level results support this: the wellness component returns a recommendation in 22 ms at the median, generated itineraries showed no overlapping visits or calendar-rule violations in scenario checks across trip lengths of 4 to 10 days, and the risk service returns predictions for 197 sites and explicit "not covered" responses for the rest. Readers are referred to the individual theses for per-class results, ablations, confusion matrices and calibration analyses.

6.4 Stakeholder and user validation

Validation of the complete platform with representative users has **not yet been carried out**. The evidence collected so far is component-level and qualitative:

- The supervisors reviewed the itinerary calendar rules and meal-plan price presentation, the SOS screens and police briefing format, and the evaluation designs of all four components.
- Reviewers of the SOS component valued that the original statement is always shown next to the AI label, and recommended keeping the 119 and 1912 numbers visible on every SOS screen, which was implemented.
- Peers who used the itinerary wizard with their own travel wishes asked for clearer price ranges and a different order of the wizard steps; both were refined.

- The wellness domain expert supplied and checked the condition–dosha–treatment mapping, durations and costs on which the wellness matching depends.

The planned integrated user acceptance test (UAT) uses the end-to-end scenarios of Table 6.3 as tasks. Participants are international or international-profile tourists for the planning, wellness and SOS tasks, and, where permission is obtained, Tourism Police officers for the dashboard tasks. Completion, time and errors are recorded per task, and each participant completes the ten-item System Usability Scale [27], interpreted with the adjective ratings of Bangor et al. [28]. Informed consent is obtained, and no personal data beyond the test session is retained. The UAT is carried out on the final integrated deployment, and its findings are presented at the final evaluation.

6.5 Achievement of common objectives

Table 6.5 maps the evidence to each group objective of Section 2.4.

Table 6.5: Achievement of the group objectives

Objective	Status	Justification and evidence
GO1 Emergency alerts	Largely achieved	Located, classified alerts routed to the nearest station with fallback and offline queue; responder model meets EQ1; incident-type model below keyword rules; latency not measured
GO2 Itineraries	Fully achieved	Calendar-valid, grounded itineraries with no overlaps in scenario checks; TC1–TC8 passed; metrics in Table 6.4
GO3 Crowd prediction	Achieved with stated limits	Better than baselines on unseen sites; target is a constructed index, so real crowding is not yet validated
GO4 Wellness matching	Fully achieved	Model and ablation results; 75/75 tests; five languages
GO5 One platform	Largely achieved	One login, one back end, one database and a home page linking all modules; one open version-pin issue; full E2E run pending
GO6 Reproducible, baseline-tested models	Fully achieved	Retraining scripts for all components; baselines and evaluation designs reported for every model
GO7 User validation	Not yet achieved	Integrated UAT planned (Section 6.4)

6.6 Integrated limitations and threats

The limitations of the integrated solution are of four kinds.

Data and validity

All four components rely, at least in part, on data that are synthetic, constructed or derived: synthetic trips (itinerary), synthetic emergency statements (SOS), a constructed crowd index (risk) and expert-rule variations of six real cases (wellness). The results therefore show how well each model learns its data, not yet how well it serves real tourists. This is the main threat to the external validity of the platform, and it is shared rather than component-specific. Several results also rest on small test sets (about 53 SOS statements, 21 dosha profiles, ten held-out sites), so a single error changes the metrics noticeably.

Integration and dependencies

- **Library versions.** The shared back end pins scikit-learn 1.5.1, while the wellness models were saved with 1.8.0 and the risk model was trained with 1.7.1. Models saved with one version may fail to load in another, so the shared requirements must be aligned before the final deployment.
- **Different site lists.** The itinerary catalogue and the crowd model use different lists of places; 142 of 350 catalogued destinations are outside the 197-site model. Crowd advice is therefore unavailable for part of a typical itinerary.
- **Partial cross-component use.** The risk itinerary check and the wellness sessions linked to the tourist identity are available, but the itinerary generator does not yet adjust plans using crowd scores or insert wellness stays automatically.
- **Shared single points of failure.** The login, database and back end are shared; a fault in them affects every component even though each component has its own fail-safe.

Operational and performance

SOS detection-to-dispatch time has not been measured, the green-site ranking takes about 7 seconds, the risk retraining endpoint is a stub, there is no live integration with police operational systems, and the platform has not been load- or penetration-tested.

Generalisability

The platform is built for Sri Lanka's registers and services. The architecture and evaluation approach generalise, but the catalogues, site list, station data and wellness centres would have to be rebuilt for another destination.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

This chapter considers the integrated platform as a product: the value it creates and for whom, the market and its alternatives, a business and sustainability model, the conditions for adoption including intellectual property, data protection and ethics, and an evaluation of its plausible contribution to the selected Sustainable Development Goals.

7.1 Value proposition and beneficiaries

Value proposition. "Plan a realistic trip from registered places, know which sites will be busy, find a suitable and safe wellness centre, and get help fast if something goes wrong — in one platform, with one login." The integrated platform creates more value than the four components separately because the tourist's identity, trip and context are shared: the SOS briefing already knows who the tourist is; a wellness stay belongs to the same trip as the itinerary; and crowd advice is attached to the places the tourist actually plans to visit.

Figure 7.1 shows the value network, and Table 7.1 lists the value created for each beneficiary with the supporting evidence.

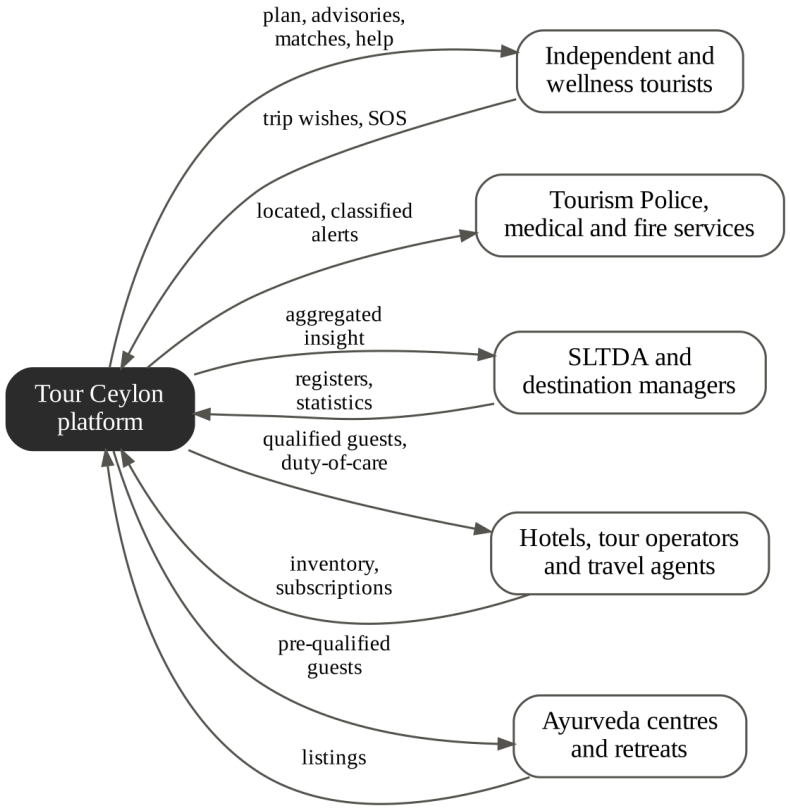


Figure 7.1: Value network of the Tour Ceylon platform

Table 7.1: Beneficiaries, value created and supporting evidence

Beneficiary	Value created	Supporting evidence
Independent tourists	Hours of manual planning replaced by a grounded, timed plan; crowd awareness; one login	Planning friction reported in practitioner sources and Table 1.1 of the itinerary thesis; itinerary and risk results
Wellness tourists	Suitable, safe centre chosen before travel, in their own language, with duration and cost	12,977 health and Ayurveda visitors staying 14.7 days on average in 2025 [1]; Wellness results
Tourists in distress	Located, structured alert to the nearest station without knowing the number or the location	Structural comparison with a voice-only call in the SOS thesis
Tourism Police and responders	Fewer clarifying questions; type, urgency, map and profile in the first message	Stakeholder review of the briefing format
SLTDA and destination managers	Visitors spread across registered and green sites; aggregated insight	Risk green-site ranking; itinerary catalogue-driven promotion
Hotels, operators, agents, wellness centres	Visibility to qualified guests matched on budget, comfort and condition; duty-of-care support	Itinerary hotel ranking; wellness matching

7.2 Market alternatives and differentiation

The target segments are, in order of priority: free independent travellers to Sri Lanka, who plan online and do not use a tour operator; wellness tourists, a smaller but long-staying and high-value segment, of whom Germany alone contributed 6,363 in 2025 [1]; and businesses and institutions that serve or protect these tourists. The global wellness tourism market alone was valued at USD 830.2 billion in 2023 [30].

Table 7.2 compares the platform with the substitutes that tourists use today.

Table 7.2: Market alternatives and the differentiation of the integrated platform

Alternative	What it does well	What it does not do	Tour Ceylon differentiation
Google Maps, TripAdvisor	Places, directions, reviews	Select, order and time visits across days; link hotels to plans	Timed multi-day plans from registered catalogues
Booking platforms	Hotel availability and price filters	Connect hotels to attractions; board-type price bands	Hotel ranking per destination with BB/HB/FB bands
Generic AI chat planners	Fluent itineraries from a prompt	Ground venues, times and prices in registers	Only registered entities; no invented venues
Hotlines 119, 1912, 1997; phone SOS features	Official, human voice response; quick trigger	Live location, profile and structured incident to the station	GPS-anchored, classified alert with briefing; offline queue
Retreat portals and online dosha quizzes	Listings and self-assessment	Constitution-aware matching to verified Sri Lankan centres with safety flags	Explained match in five languages with safety screening
Official annual site statistics	Authoritative annual totals	Daily, site-level estimates	Daily crowd score and risk class per site

The defensible differentiation is the combination: no reviewed alternative grounds planning, crowd context, wellness matching and emergency response in Sri Lankan data under one identity. Each component's curated dataset (catalogues with meal-plan prices, the defect-audited 197-site dataset, the labelled emergency corpus, the verified wellness centres) is also costly for a competitor to reproduce.

7.3 Business model and feasibility

A business model was outlined using the structure of the Business Model Canvas [31]. Its central rule is that safety is never paywalled: the SOS service is free for

every tourist, and revenue comes from optional premium features for tourists and from business and institutional partners (Table 7.3).

Table 7.3: Candidate revenue and sustainability streams (not yet validated with customers)

Stream	Payer	Description
Free core	–	Itinerary generation, crowd advisories, wellness matching and SOS free for tourists, to build trust and adoption
Premium planning	Tourists	Branded exports, alternative plans and saved trip histories
Planning API subscription	Travel agents and tourism websites	Bulk itinerary generation and hotel ranking through the API
Wellness verified listing and enquiry commission	Ayurveda centres and retreats	Verified-listing badge for centres that pass verification; commission on enquiries that convert into stays
Duty-of-care subscription	Hotels and tour operators	Monitoring and support of guests' emergencies through the SOS dashboard
Institutional partnership	SLTDA, Tourism Police	Hosted SOS and crowd-insight service as a digital complement to the 1912 hotline

Channels and partners. Tourists are reached through the platform website, hotels, tour operators and tourism-authority channels; key partners are SLTDA, the Tourism Police, registered accommodation and wellness providers, and the domain experts who maintain the catalogues. **Resources and costs.** The main resources are the curated datasets, the trained models and the shared platform. Costs are hosting, SMS and translation usage, data maintenance and support. Because all models are light Random Forest and linear models that run on a standard server and load once at start-up, the marginal cost of a request is low.

Feasibility. Technical feasibility is supported by the working integrated prototype and by the measured latency of the components (for example 22 ms for a wellness recommendation). Operational feasibility depends on keeping catalogues current, which the retraining scripts make a data task rather than a programming task. Financial feasibility has not been established: no cost data or customer commitments exist, so no financial forecast is presented, and the revenue streams above are hypotheses to be tested with partners.

7.4 Adoption intellectual property and compliance

Adoption barriers

- **Trust.** Tourists and officers must trust AI suggestions. The platform shows the reasons behind each recommendation, the confidence of each SOS label next to the original statement, and plain-language statements of each model's accuracy.
- **Institutional adoption.** Police and tourism authorities adopt new channels slowly. A phased pilot (consultation, a pilot in one or two areas, technical integration, field tests with simulated alerts, phased expansion) is proposed by the SOS component owner; letters connected with Police Headquarters and immigration authorities, as reported by the SOS component owner, must not be read as approval for live deployment.
- **Data freshness.** Catalogues, stations and centres change; out-of-date data would undermine the grounding principle.

Intellectual property and licensing

The code, curated datasets, training pipelines, briefing design and integration code are the intellectual property of the project team, subject to SLIIT's regulations on student research. Data compiled from third-party sources (public listings and reviews, official statistics and registers) must be used within the terms of those sources, and any commercial use would require a review of those terms and, where needed, licences or partnerships with the data owners.

Data protection, ethics and safety

- **Personal data.** The platform processes identity, contact details, location, health information and descriptions of incidents. Processing must follow the Personal Data Protection Act, No. 9 of 2022 [25]: minimum data, consent at the point of use (location at SOS time, health answers for matching), purpose limitation, retention limits and secure storage. A data-protection impact assessment is required before live deployment, together with a review of cross-border processing by speech and translation services.
- **Health information.** Wellness matching is decision support, not medical advice; the interface states this and directs tourists to qualified practitioners.

- **Safety-critical AI.** The SOS label is advisory; delivery of the alert never depends on it, and the official numbers 119 and 1912 remain visible. False reports are discouraged and duplicate alerts controlled.
- **Honest estimates.** Crowd scores are presented as estimates of a constructed index, not as observed crowding.
- **Fairness.** English-centric models may serve some tourists better than others; the wellness interface supports five languages and the SOS voice path translates input, but per-language performance must be monitored.
- **Academic integrity.** Generative AI assistance is disclosed in each individual thesis and in the declaration of this report.

Professional responsibility

The team followed professional practice in recording the evaluation design of every result, disclosing data defects and weak results (for example the fourteen-class SOS result and the optimistic itinerary AUC), separating implemented from planned features, and keeping secrets out of the code.

7.5 Sustainable Development Goal contribution

Table 7.4 evaluates the plausible contribution of the platform to the SDG targets selected in Section 1.3 [12], the possible negative effects and the measures needed to sustain the benefits.

Table 7.4: Evaluation of the plausible SDG contribution, risks and sustaining measures

SDG	Plausible contribution	Possible negative effect	Sustaining measure
3 Health and well-being	Faster, better-informed medical response; safer choice of traditional care	False confidence in an unvalidated system; misleading health matches	Field pilot with responders; practitioner review of mappings; clear disclaimers
8.9 Sustainable tourism	Longer, better-planned stays spread across registered providers in many districts	Benefits concentrated on businesses already listed	Open, verified listing process; regular catalogue updates
9 Innovation and infrastructure	Reusable, data-grounded digital services for the sector	Dependence on a small team for maintenance	Documentation and one-click retraining; institutional partner

SDG	Plausible contribution	Possible negative effect	Sustaining measure
11 Safer communities	Located, structured emergency reporting; crowd-aware visiting	Misuse of location data; surveillance concerns	Consent at SOS time; role-based access; retention limits
12.b Monitoring sustainable tourism	Promotion of less-crowded and green sites through data	Redirecting visitors to fragile nature sites, where the crowd model is least accurate	Use site capacity limits; validate nature-site predictions with operators before promotion

The contribution is plausible rather than demonstrated. It depends on adoption by tourists and institutions, on the accuracy of models that have so far been evaluated on synthetic or constructed data, and on responsible data handling. The negative effect noted for SDG 12.b is specific to the integrated platform: the crowd model's largest errors occur at nature sites such as national parks, which are also the sites a green-site ranking might promote, so promotion should be bounded by site capacity and checked with site operators.

8 CONCLUSION AND FUTURE DIRECTIONS

This chapter answers the common problem using the integrated evidence, states the principal contribution of the team beyond the four components, and sets out the priorities for future work.

8.1 Overall conclusion

The common problem was that international tourists in Sri Lanka have no single, trustworthy digital service for planning a feasible trip, judging how busy sites will be, choosing suitable wellness care and getting help quickly in an emergency. The integrated evidence shows that such a service can be built with established web technologies and lightweight, explainable machine learning, and that it can be grounded in Sri Lankan data. Tour Ceylon runs as one platform with one login, one back end and one database; each of its four components makes one decision that the others rely on; and each component remains usable when its AI part or the network fails.

Where the evaluation was strict, the AI components improved on simple alternatives: crowd prediction on unseen sites reduced the error of the best baseline by about 40%, wellness matching raised the quality and budget fit of the top recommendation substantially, and emergency routing directed about four in five held-out statements to the correct service. The evaluation also showed where the solution is not yet good enough: the served fourteen-class SOS model is weaker than keyword rules, near-perfect itinerary scores reflect the synthetic labelling, crowd predictions are least reliable at nature sites, and no component has yet been validated with real tourists. The answer to the common problem is therefore positive for feasibility and integration, and open for real-world effectiveness.

8.2 Principal integrated contribution

- **An integrated, data-grounded tourism platform for Sri Lanka** in which planning, crowd context, wellness matching and emergency response share one identity and one data layer, with documented interfaces (Appendix A) that allow each component to be maintained or replaced independently.

- **A shared fail-safe design** in which every AI decision is advisory and every component has a defined degraded mode, so that the failure of a model never blocks a plan, a match or an alert.
- **A shared standard of honest evaluation**: every model evaluated against simple baselines with its evaluation design stated, grouped or unseen-data results treated as primary, and weak or optimistic results reported openly. The team learned that this standard changed decisions, for example which SOS model to serve and how to interpret itinerary scores.
- **Reproducible, data-driven maintenance**: catalogues, sites, corpora and centres can be updated and the models retrained by script without code changes.

8.3 Future directions

Future work is prioritised by the strength of evidence it would add and the risk it would remove.

1. **Complete the integrated validation.** Align the scikit-learn version in the shared environment, run the end-to-end scenarios of Table 6.3, and carry out the integrated UAT with tourists and, with permission, Tourism Police officers, reporting task completion and SUS scores.
2. **Replace synthetic and constructed data with real data.** Collect consented real trip inputs and edits (itinerary), consented and de-identified real emergency statements including Sinhala, Tamil and code-mixed input (SOS), observed visitor counts from site operators (risk) and real treatment outcomes from partner centres (wellness).
3. **Deepen the integration.** Let the itinerary generator call the risk itinerary check to move visits away from predicted High-risk days, insert matched wellness stays into the itinerary, align the itinerary and risk site lists, and give the SOS briefing access to the tourist's current itinerary.
4. **Improve the weakest models.** Serve Logistic Regression or an ensemble for SOS incident type, investigate the nature-site offsets in the crowd model, and expand the minority classes of the wellness safety classifier.
5. **Prepare for deployment.** Add TLS, multi-factor authentication for officers, an audit log, a data-retention schedule and a data-protection impact assessment;

measure SOS detection-to-dispatch time; load-test the unified back end; and pilot the SOS service with the Tourism Police in one area.

6. **Test the business model.** Validate the revenue hypotheses of Table 7.3 with hotels, agents and wellness centres before any financial forecast is made.

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APPENDICES

Appendix A Integrated interface specification

This appendix specifies the interfaces through which the components are reached in the unified back end. All endpoints accept and return JSON unless stated otherwise. Paths are those documented in the individual theses; representative paths for the SOS component should be confirmed against the final code.

A.1 Shared platform

Table A.1: Shared platform interfaces

Method and path	Purpose	Request	Response
Health check	Status of all components	–	Status of travel, SOS, wellness and risk
POST /api/tourists/token-exchange	Open SOS from the itinerary client without a second login	Platform access token	SOS session token

A.2 SOS Emergency Safety System

Table A.2: SOS interfaces (representative paths)

Method and path	Purpose	Request	Response
POST /api/incidents	Create SOS	GPS, statement, optional journey points	Incident with type, confidence, urgency, responder, station
GET /api/incidents/mine	Tourist's own incidents	–	Incidents and statuses
GET /api/police/incidents	Incidents for the operator's station	–	Incident list with map data
POST /api/police/incidents/{id}/ack, /escalate, /close	Lifecycle actions	–	Updated status
POST /api/incidents/{id}/chat; /location	Chat; live GPS update	Text or voice; coordinates	Message; location
GET /api/services/nearby	Nearest police and medical services	Coordinates	Services with distance

Method and path	Purpose	Request	Response
SocketIO: incident:new, incident:status, incident:location, chat:message	Real-time events	–	Event payloads per station room

A.3 AI Itinerary Generator

Table A.3: Itinerary interfaces

Function	Service	Request	Response
Attraction suggestion	`suggest_from_trip()`	Trip input: dates, experience text, budget, comfort, meal plan	Ranked catalogue attractions
Hotel ranking	`ranked_hotels_for_destination()`	Destination, budget, comfort, meal plan	Ranked hotels with nightly and stay price bands
Itinerary generation	`generate()` via `build_itinerary_payload()`	Trip input and selections	Ordered rows: arrival, check-in, timed visits, departure; HTTP 503 if the bundle is missing

A.4 Tourism Risk and Context Intelligence System

Table A.4: Risk interfaces

Path	Purpose	Request	Response
/api/risk (prediction)	Daily crowd score and risk class	Site and date	Score 0–1; Low (< 0.45), Medium (≥ 0.45) or High (≥ 0.75); or "not covered"
/api/risk/itinerary-check	Risk check for a planned itinerary	List of site visits	Score and class per visit
/api/risk/green-sites	Ranking of eco-sites	Date or period	114 eco-sites ranked (about 7 s)

A.5 Wellness Component

Table A.5: Wellness interfaces

Method and path	Purpose	Request	Response
GET /api/conditions?path=	Condition or retreat types	ayurveda or spiritual	List of types

Method and path	Purpose	Request	Response
GET /api/centers?path=	Centres for a path	path	Names, categories, districts
POST /api/recommend	Full recommendation	Condition, age, gender, four lifestyle answers, budget tier, top_n	Dosha and confidence, duration, outcome, ranked centres with score breakdown and risk flags
POST /api/match	Ranking for a known dosha	Condition, dosha, budget tier	Ranked centres
GET /api/center/<name>	Centre detail	Centre name	Detail, flags, offers
POST /api/cost-estimate	Duration and cost	Condition, price tier, travellers	Minimum and maximum days and cost
POST /api/classify-symptom	Free-text interpretation	Text, language, domain	Category, confidence, model accuracy, no-signal flag
POST /api/safety-check	Review safety screening	List of review texts	Keyword and ML flags with confidence
POST /api/predict-outcome	Outcome estimate	Age, dosha, condition, duration	Predicted improvement (%)

A.6 Shared database tables

Table A.6: Database tables that cross component boundaries

Table or entity	Owner	Key fields	Read by
Tourists / users	Platform	tourist_id, role, profile	All components
UserTripInput	Itinerary	trip id, dates, experience text, selections	Platform; other components
Attraction, Accommodation	Itinerary	name, destination, slots; BB/HB/FB min–max	Itinerary
Incidents, stations, chat, journey points	SOS	incident id, tourist_id, station, status	Police and immigration dashboards
wellness_centers, wellness_reviews, wellness_condition_mappings	Wellness	centre name, district, category, verification	Wellness
wellness_matching_sessions	Wellness	session id, tourist_id, condition, result	Platform; Itinerary (planned)

Appendix B Team contribution declaration

The members of group R26-IT-152 agree that the individual and shared contributions are as recorded in Table 4.5 of Section 4.5 and summarised below. Each member is the sole owner of their component's research, data, models and component code, as reported in their individual thesis. The shared problem definition, architecture, integration and this common report are joint work.

Table B.1: Contribution declaration

Member	Student ID	Individual component	Shared contribution	Confirmed (signature and date)
De Silva D.S.K	IT22108654	SOS Emergency Safety System	Token exchange, shared tourist identity in briefings	
Wanniarachchi P.M.R	IT22225092	AI Itinerary Generator	Trip model, catalogue seeding, itinerary client as platform entry point	
Abinaya Rajasekara	IT22090058	Tourism Risk and Context Intelligence System	Risk mount in the unified back end; risk API for other clients	
De Silva D.C.M	IT22108586	Wellness Component	Wellness blueprint and migration; integration version report; common report drafting	